QQ-P-416D
20 November 1985
SUPERSEDING
QQ-P-416C
29 January 1971

FEDERAL SPECIFICATION

PLATING, CADMIUM (ELECTRODEPOSITED)

This specification is approved by the Commissioner, Federal Supply Service, General Services Adminstration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for electrodeposited cadmium plating.

1.2 Classification. Cadmium plating shall be of the following types and classes, as specified (see 6.2):

1.2.1 Types.

Type	Description
I	As plated
II	With supplementary chromate treatment (see 3.3.9)
III	With supplementary phosphate treatment (see 3.3.10)

1.2.2 Classes.

Class	Thickness, inch,
1	0.00050
2	0.00030
3	0.00020

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitations for offers, form a part of this specification to the extent specified herein.

Federal Specifications

TT-C-490 - Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings

AREA MFFP

NO DELIVERABLE DATA REQUIRED BY THIS DOCUMENT

DISTRIBUTION STATEMENT A, Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

Federal Standards

FED-STD-151	-	Metals; Test Methods
FED-STD-595	-	Colors

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-S-5000	-	Steel, Chrome-Nickel-Molybdenum, (E4340), Bars and Reforging Stock
MIL-S-5002	-	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems

Military Standards

MIL-STD-105	-	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-1312	-	Fasteners, Test Methods

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bid or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

American Society for Testing and Materials (ASTM) Standards

B117 - Salt Spray (Fog) Testing
B244 - Thickness of Anodic Coatings on Aluminum and of other Non-conductive Coatings on Nonmagnetic Basis Metals with Eddy Current Instruments, Measurement of
B487 - Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section, Measurement of
B499 - Coating Thicknesses by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metals, Measurement of
B504 - Thickness of Metallic Coatings by the Coulometric Method, Measurement of
B567 - Coating Thickness by the Beta Backscatter Method, Measurement of
B568 - Coating Thickness by X-Ray Spectrometry, Measurement of
E8 - Tension Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preproduction. Unless otherwise specified (see 6.2), the contractor shall furnish sample units for inspection and approval (see 4.4). Preproduction hydrogen embrittlement tests are not required for steels heat treated below 160 ksi.

3.2 Materials. The materials used shall be such as to produce platings which meet the requirements of this specification.

3.3 General requirements.

3.3.1 High tensile strength steel parts. Except for fasteners, all steel parts having an ultimate tensile strength greater than 220 ksi (or hardness greater then RC 46) shall not be plated without specific approval of the acquiring activity. Steel fasteners having a hardness of RC 43 or higher shall not be plated without specific approval of the acquiring activity.

3.3.2 Stress relief treatment. Unless otherwise specified, parts which are machined, ground, cold formed or cold straightened, after heat treatment, shall receive a stress relief treatment in accordance with Table I prior to shot peening, cleaning, and plating for the relief of damaging residual tensile stresses (see 6.2). For fastener hardware, the results of cold working of the head-to-shank fillet radius, and thread rolling after heat treat, when required, are not damaging residual stresses. Stress relief shall be the last operation before cleaning and plating. If the part is to be shot peened, it should be done after stress relief and before cleaning. No further machining, grinding or cold straightening shall be performed after stress relieving.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

3.3.3 Cleaning. Unless otherwise specified, all steel parts shall be cleaned in accordance with paragraph 3.4 of MIL-S-5002 (see 6.2).

3.3.4 Plating application. Unless otherwise specified, the plating shall be applied after all basis metal heat treatments and mechanical operations, such as machining, brazing, welding, forming and perforating of the article, have been completed (see 6.2).

3.3.5 Underplating. Unless otherwise specified, cadmium shall be deposited directly on the basis metal without a preliminary plating of other metal, such as nickel or copper, except in the case of parts made of corrosion resistant alloys on which a preliminary plating of nickel is permissible or of parts made of aluminum on which a preliminary treatment, such as the zincate process, is permissible (see 6.2).

3.3.6 Coverage. Unless otherwise specified, the plating shall cover all surfaces as stated in 3.4.1, 3.4.1.1 and 3.4.1.2 including roots of threads, corners and recesses (see 6.2).

3.3.7 Luster. Unless otherwise specified, either a bright or dull luster shall be acceptable (see 6.2). However, the use of brightening agents in the plating solution is prohibited on components with an ultimate tensile strength greater than 160 Ksi (or hardness greater than 36 Rc).

3.3.8 Hydrogen embrittlement relief. Unless otherwise specified in the contract or order or stated in the end product specifications, all parts shall be baked according to Table I. Parts required by Table I to be baked when tested in accordance with 4.6.4, shall not crack or fail by fracture (see 4.5.2.3.4). Plated springs and other parts subject to flexure shall not be flexed prior to hydrogen embrittlement relief treatment. In the case of types II and III treated parts, the baking treatment shall be done prior to the application of the supplementary coatings. Cadmium plated surfaces passivated as a result of the baking operation shall be reactivated prior to receiving the type II supplementary chromate treatment (see 6.7).

3.3.9 Chromate treatment (type II). Unless otherwise specified in the contract or order (see 6.2), the chromate treatment required for conversion to type II shall be a treatment in or with an aqueous solution of salts, acids, or both, to produce a continuous smooth, distinct protective film, distinctly colored iridescent bronze to brown including olive drab and yellow, and forest green. The articles so treated shall be thoroughly rinsed and dried in accordance with the requirements of the process used. Type II plating shall be similar in appearance to platings on separate specimens which are capable of passing the salt-spray test (see 3.4.4, 4.5.2.3.3, and 4.6.3). Usual chromic and nitric acid bright dips for cadmium are not chromate treatments. Type II, class 3 coatings shall meet the minimum thickness requirement after Type II treatment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

3.3.10 Phosphate treatment (type III). Unless otherwise specified in contract or order, the phosphate treatment required for conversion to type III shall produce a tightly adherent film conforming to type I of TT-C-490 (see 6.2).

3.4 Detail requirements.

3.4.1 Thickness of plating. Unless otherwise specified in the contract or order (see 6.2), the thickness of cadmium for other than fastener hardware (see 3.4.1.1) shall be as specified in table II on all visible surfaces which can be touched by a ball 0.75 inch in diameter. Where class 1 is specified, all other visible surfaces shall be class 2 minimum thickness. If the maximum thickness for class 1 is not specified in the contract, order or applicable drawing, the thickness shall not exceed 0.0008 inch (0.8 mil). Where class 2 is specified, all other visible surfaces shall be class 3 minimum thickness. Where class 3 is specified, all other visible surfaces shall be not less than 0.00015 inch minimum thickness.

3.4.1.1 For aerospace fastener hardware. Unless otherwise specified in the product specifications, aerospace fastener hardware shall have Class 2 thickness (see 4.5.1). No bare spots shall be present.

3.4.1.1.1 For Army use. Unless otherwise specified in the product specification, for Army use, fastener hardware shall have Class 3 plating (see 4.5.1). No bare spots shall be present.

3.4.1.1.2 Stress durability. Embrittlement testing of aerospace fasteners shall be as specified in the applicable fastener procurement specification.

3.4.1.2 For other than fastener hardware. For other than fastener hardware, the cadmium plating shall be class 1 thickness unless otherwise specified in the contract or order or controlled by the following exceptions (see 6.2):

(a) Articles with portions externally threaded shall have a minimum of class 2 thickness on the threaded portions
(b) Holes and other openings and articles with internal threads from which the external environment is completely excluded shall not be subjected to thickness requirements but shall show evidence of coating. There shall be no bare areas. In special cases, such as deep small diameter holes (holes with a depth equal to or greater than one diameter), the requirement for no bare areas should be waived and the method of protection specified in the contract or order.

3.4.2 Types. Unless otherwise specified in the contract or order (see 6.2), the cadmium plating shall be type II, except Type III may be used on surfaces to be painted. (see 6.1.2 and 6.1.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

3.4.3 Adhesion. The adhesion of the plating shall be such that when examined at a magnification of 4 to 10 diameters, the plating shall not show separation from the basis metal nor from any underplating at the interface, nor shall any underplate show separation from the basis metal at the interface when subjected to the tests described in 4.6.2. The interface between the underplate and the basis metal is the surface before plating. The formation of cracks in the plating caused by rupture of the basis metal, the underplate or combination of both which do not result in flaking, peeling or blistering of the plating shall not be considered as nonconformance to this requirement.

3.4.4 Corrosion resistance. Cadmium plating with the type I plating shall show no red rust. Type II shall show neither white corrosion products of cadmium nor basis metal corrosion products at the end of 96 hours when tested by continuous exposure to the salt spray in accordance with 4.6.3. The appearance of corrosion products visible to the unaided eye shall be cause for rejection, except that white corrosion products at the edges of specimens shall not constitute failure.

3.5 Workmanship.

3.5.1 Basis metal. The basis metal shall be free from visible defects that will be detrimental to the appearance or protective value of the plating. The basis metal shall be subject to such cleaning and plating procedures as necessary to yield platings herein specified.

3.5.2 Plating. The cadmium plating shall be smooth, adherent, uniform in appearance, free from blisters, pits, nodules, burning and other defects. The plating shall show no indication of contamination or improper operation of equipment used to produce the cadmium deposit, such as excessively powdered or darkened plating. Superficial staining, which has been demonstrated as resulting from rinsing, or slight discoloration resulting from any drying or baking operations as specified (see 3.3.8) shall not be cause for rejection. All details of workmanship shall conform to the best practice for quality plating.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.4)
(b) Quality conformance inspection (see 4.5)

4.3 Inspection conditions. All inspections shall be performed in accordance with the test conditions specified in the applicable test method or applicable paragraph in this specification.

4.4 Preproduction tests. Before commencing production, the processor shall perform tests on specimens (see 4.5.3) for hydrogen embrittlement relief (see 3.3.8, 4.4.1 and 4.6.4), corrosion resistance (see 4.6.3), thickness (see 4.6.1) and adhesion (see 4.6.2). After successful completion of the above tests, production parts may be plated. Complete report of the tests shall be permanently kept on file by the processor and made available to any purchaser upon request.

4.4.1 Preproduction samples. Unless otherwise specified in the contract or purchase order, as soon as practicable after award of the contract or order, the contractor (processor) shall demonstrate that the process that shall be used shall result in freedom from hydrogen embrittlement damage by preparing and testing specimens for conformance to the requirements of this specification and MIL-S-5002 for production process control. The processor shall prepare four round notched steel specimens, as specified in 4.5.3.3, from two individual heats for a total of 8 specimens (see 6.2.2). The specimens shall be the same steel alloy as that which is to be plated on a production basis. The specimens shall be heat treated to the maximum tensile strength representing production usage. The specimens shall be given the same pre-treatments, electroplating and post-plating treatments proposed for production. The specimens shall be subject to test detailed in 4.6.4. The process shall be considered satisfactory if there is no indication of failure or cracking of the plating on any of the specimens. The test results and production control information shall be forwarded to the contracting officer for approval. Until approval has been received, the parts shall not be plated. When a processor is in continuous production of plating from contract to contract, preproduction inspection may be waived at the discretion of the procuring activity (see 6.2).

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the following:

(a) Process control inspection
(b) Lot sampling inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

4.5.1 Process control inspection.

4.5.1.1 Control records. The processor shall maintain a record of the history of each processing bath, showing all additions of chemicals or treatment solutions to the unit, the results of all chemical analyses performed and the quantity of parts plated during operation. Upon request of the acquisition activity, such records, as well as reports of the test results, shall be made available. These records shall be maintained for not less than one year after completion of the contract or purchase order.

4.5.1.2 Analytical inspection. The equipment, procedures and operations employed by the processor shall be capable of producing high quality electrodeposited cadmium plating as specified in this document. Analytical inspection and additions shall be conducted to maintain the process solutions within established operating ranges.

4.5.1.3 Process control tests.

4.5.1.3.1 Frequency of tests. Unless otherwise specified in the contract or purchase order, process control tests shall be conducted to assure continuous control of the process, as required by MIL-S-5002, and to minimize the possible occurrence and acceptance of items with detrimental hydrogen embrittlement during production. Process control specimens shall be prepared and tested in accordance with table III. The tested specimens shall conform to the requirements of this specification. These tests shall be conducted at least once a month or more frequently if required by the acquisition activity. Lot tests can be substituted for weekly or monthly tests. These tests are conducted to determine conformance of the electrodeposited platings with the requirements of this specification and are acceptable as evidence of the properties being obtained with the equipment and procedures employed. Copies of these certified test results will be made available to the government buying agency upon request.

4.5.1.3.2 Test specimens. The test specimens for the process control inspection shall be prepared in accordance with 4.5.3 through 4.5.3.3, as applicable, for the thickness, adhesion, corrosion resistance and hydrogen embrittlement relief tests specified in table III. Specimens for the process control embrittlement relief tests shall be four round notched steel specimens, one or more heats of alloy steel 4340 conforming to MIL-S-5000, and shall be heat treated to the maximum tensile strength.

4.5.2 Lot sampling inspection. The lot sampling inspection shall consist of the examinations and tests specified in 4.5.2.2 through 4.5.2.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

4.5.2.1 Lot. A lot shall consist of plated articles of the same basis metal composition, class and type plated and treated under the same conditions and approximately the same size submitted for inspection at one time.

4.5.2.2 Sampling for visual examination and nondestructive tests. Sampling for visual examination and nondestructive tests shall be conducted, in accordance with MIL-STD-105 or table IV, at the option of the supplier. A sample of plated parts or articles, except for those barrel plated, shall be taken at random from each lot, the number of articles in accordance with MIL-STD-105, inspection level II with an acceptable quality level (AQL) of 1.5 percent defective, or as indicated in table IV. Barrel plated parts or articles shall be sampled in accordance with inspection level S-3 of MIL-STD-105 with an AQL of 4.0 percent defective. The lot shall be accepted or rejected according to the procedures in 4.5.2.2.1 for visual examination and 4.5.2.2.2 for plating thickness (nondestructive tests).

4.5.2.2.1 Visual examination. Samples selected in accordance with 4.5.2.2 shall be examined for compliance with requirements of 3.5.2 after plating. If the number of nonconforming items exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.5.2.2.2 Thickness of plating (nondestructive test). Samples selected in accordance with 4.5.2.2 shall be inspected and the plating thickness measured by the applicable test detailed in 4.6.1 at several locations on each article as defined in 3.4.1.1 or 3.4.1.2, as applicable, for compliance with the requirements. Measurements on fastener hardware (see 3.4.1.1) shall be made at locations defined in MIL-STD-1312, test 12. The part or article shall be considered nonconforming if one or more measurements fail to meet the specified minimum thickness. If the number of defective items in any sample exceeds the acceptance number for the specified sample, the lot represented by the sample shall be rejected. Separate specimens (see 4.5.3.1) shall not be used for thickness measurements unless testing of parts is not possible.

4.5.2.3 Sampling for destructive tests. A random sample of four plated parts or articles shall be taken from each lot for each destructive test or separately plated specimens shall be prepared in accordance with 4.5.3 through 4.5.3.3 to represent each lot. If the number of articles in the lot is four or less, the number of articles in the sample shall be specified by the procuring activity (see 6.2).

4.5.2.3.1 Thickness of plating (destructive tests). For sampling and testing for thickness of plating by destructive testing. Samples selected in accordance with 4.5.2.3 shall be measured for plating thickness by the applicable test detailed in 4.6.1, at several locations on each article as defined in 3.4.1, 3.4.1.1 or 3.4.1.2 for compliance with the requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

Measurements on fastener hardware (see 3.4.1.1) shall be made at locations defined in MIL-STD-1312, test 12. If the plating thickness at any point on any article or specimen is less than the specified minimum thickness, the lot shall be rejected. Separate specimens (see 4.5.3.1) shall not be used for thickness measurements, unless a need has been demonstrated.

4.5.2.3.2 Adhesion (destructive tests). The adhesion test shall be performed on each lot in accordance with 4.6.2. The articles or specimens used for the destructive thickness test (see 4.5.3.1), if of suitable size and form, may be used as the test pieces for the adhesion test to determine compliance with the requirements of 3.4.3. Failure of one or more of the test pieces shall constitute failure of the lot.

4.5.2.3.3 Corrosion resistance (destructive tests). When specified in the contract or order, compliance with the requirements for corrosion resistance of type II treated articles shall be determined (see 6.2). A set of four separate test specimens, prepared in accordance with 4.5.3 and 4.5.3.2 in lieu of the treated coated articles, shall be used to determine compliance with the requirements for corrosion resistance (see 3.4.4). Failure of one or more of the test specimens shall reject the lot.

4.5.2.3.4 Hydrogen embrittlement relief (destructive tests). Unless otherwise specified in the contract or order (see 6.2), conformance to the requirements of 3.3.8 for hydrogen embrittlement relief of treated parts shall be determined. A random sample of four plated articles shall be taken from each lot or four specimens, prepared in accordance with 4.5.3 and 4.5.3.3, shall be used to represent the lot. When tested as specified in 4.6.4, cracks or failure by fracture shall be cause for rejection. Failure of one or more of the test pieces shall reject the lot.

4.5.3 Specimen preparation. When the plated articles are of such form, shape, size and value as to prohibit use thereof, or are not readily adaptable to a test specified herein, or when destructive tests of small lot sizes are required, the test shall be made by the use of separate specimens plated concurrently with the articles represented. The separate specimens shall be of a basis metal equivalent to that of the articles represented. "Equivalent" basis metal includes chemical composition, grade, condition and finish of surface, prior to plating. For example, a cold-rolled steel surface should not be used to represent a hot-rolled steel surface. Due to the impracticality of forging or casting separate test specimens, hot-rolled steel specimens may be also used to represent forged and cast-steel articles. The separate specimens may be also cut from scrap castings when ferrous alloy castings are being plated. These separate specimens shall be introduced into a lot at regular intervals prior to the cleaning operations, preliminary to plating and shall not be separated therefrom until after completion of plating.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

Conditions affecting the plating of specimens including the spacing and plating media, bath agitation, temperature, etc., in respect to other objects being plated shall correspond as nearly as possible to those affecting the significant surfaces of the articles represented. One plating bath can not be used to substitute for similar plating baths; the tests must be performed on each plating bath. Separate specimens shall not be used for thickness measurements, unless testing of parts is not possible.

4.5.3.1 Specimens for thickness and adhesion tests. If separate specimens for thickness and adhesion tests are required, they shall be strips approximately 1 inch wide, 4 inches long and 0.04 inch thick.

4.5.3.2 Specimens for corrosion resistance tests. If separate specimens for corrosion resistance test are required, they shall be panels not less than 6 inches in length, 4 inches in width and approximately 0.04 inch thick.

4.5.3.3 Specimens for embrittlement relief. Separate specimens for the embrittlement relief test shall be round notched specimens with the axis of the specimen (load direction) perpendicular to the short transverse grain flow direction. The configuration shall be in accordance with figure 8 of ASTM E8 for rounded specimens. Specimens shall have a 60 degree V-notch located approximately at the center of the gage length. The cross section area at the root of the vee shall be approximately equal to half the area of the full cross section area of the specimen's reduced section. The vee shall have a 0.010 $\pm$ 0.0005 radius of curvature at the base of the notch (see 6.2.2). Specimens plated with the part lot to represent the parts for testing purposes shall be prepared so that the load is applied in the longitudinal grain direction.

4.6 Tests.

4.6.1 Thickness. For nondestructive measuring of plating thickness, procedures in accordance with FED-STD-151, method 520 (electronic test), ASTM B499 (magnetic test), ASTM B244 (eddy current), ASTM B567 (test by beta radiation backscatter principle) or ASTM B568 (X-ray spectrometry) may be used. For destructive measuring of plating thickness, procedures in accordance with ASTM B487 (microscopic) of ASTM B504 (coulometric) may be used. In addition to the above, the other procedures embodied in MIL-STD-1312, test 12 may be used for thickness measurement of plating fastener hardware. Thickness measurements of cadmium platings, types II and III, shall be made after application of the supplementary treatments. When the coulometric test is used, the supplementary treatment shall be removed prior to testing. The chromate film may be removed from the type II coating by using a very mild abrasive (a paste of levigated alumina rubbed on with the finger). The phosphate coating may be removed from the type III coating by immersing the specimen in a 10 percent solution of NaOH and scrubbing with a rubber policeman (usually takes from 10 to 15 minutes).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

4.6.2 Adhesion. Adhesion may be determined by scraping the surface or shearing with a sharp edge, knife or razor through the plating to the basis metal and examining at four to ten diameters magnification for evidence of non- adhesion. Alternately, the article or specimen may be clamped in a vise and the projecting portion bent back and forth until rupture occurs. If the edge of the ruptured plating can be peeled back or if separation between the plating and basis metal can be seen at the point of rupture when examined at four diameter magnification, adhesion is not satisfactory.

4.6.3 Corrosion resistance. Corrosion resistance shall be conducted in accordance with ASTM B117 (salt spray test) for 96 hours or in accordance with MIL-STD-1312, test 1 for fastener hardware. To secure uniformity of results, type II supplementary coatings shall be aged at room temperature for 24 hours minimum before subjection to the salt spray.

4.6.4 Embrittlement relief. Compliance with 3.3.8 shall be determined with samples of plated parts or specimens plated in accordance with 4.5.2.3.4. Parts such as spring pins, lock rings, etc., which are installed in holes or rods, shall be similarly assembled using the applicable parts specifications or drawing tolerances which impose the maximum sustained tensile load on the plated part. The selected samples shall be subjected to a sustained tensile load equal to 75 percent of the rated ultimate load. Fastener hardware shall be tested in accordance with MIL-STD-1312, test 5 or test 14, as applicable. Parts which require special fixtures or extreme loads to comply with the above requirements, may be represented by separate specimens prepared in accordance with 4.5.3.3. The notched specimens shall be subjected to a sustained tensile load equal to 75 percent of the ultimate notch strength of the material. The articles, parts or specimens shall be held under load for not less than 150 hours to RC 36 and higher, and then be examined for cracks or fractures.

4.6.4.1 Stress durability. Fasteners shall be subject to the stress durability test in accordance with MIL-STD-1312 Test 5 or Test 14, as applicable. The tensile load shall be equal to 75 percent of the rated ultimate for not less than 40 hours and then examined for cracks or fractures. Parts which require special fixtures to comply with the above requirements may be represented by specimens (in lieu of the fasteners) plated with the lot and tested in accordance with 4.5.3.3.

4.6.4.2 Notched specimens. Notched specimens plated with the parts lot shall be subjected to a sustained tensile load equal to 75 percent of the ultimate tensile strength of the material. The specimens, RC-36 and higher, shall be subjected to the load for not less than 200 hours and then examined for cracks and fractures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

This section is not applicable to this specification.

6. NOTES

6.1 Intended use.

6.1.1 General usage. The electrodeposited cadmium platings covered by this specification are intended for use as corrosion protective coating on ferrous parts. On ferrous parts heat-treated to or having ultimate tensile strength greater than 160 ksi (approximately Rockwell hardness of C36), processing used for cleaning and cadmium deposition should be such that no hydrogen embrittlement will result. On steel heat-treated or having an ultimate strength greater than 240,000 psi, use of cadmium electrodeposition should be reviewed to ensure satisfactory performance.

6.1.2 Type II treatment. The prime purpose of chromate finishes (type II) on electrodeposited cadmium platings is to retard or prevent the formation of white corrosion products on surfaces exposed to stagnant water, high humidity atmospheres, salt water, marine atmospheres or cyclic condensation and drying. Some types of chromate films have proved satisfactory as a base for paints.

6.1.3 Type III treatment. The prime purpose of phosphate finishes (type III) on electrodeposited cadmium platings is to form a paint base.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Type and class required (see 1.2.1, 1.2.2, 3.4.1, and 3.4.2)
(c) When plating is to be applied, if other then specified (see 3.3.1, 3.3.2, 3.3.4 and 3.3.6)
(d) Stress relief treatment, if other then specified (see 3.3.2)
(e) Cleaning of steel, if other than specified (see 3.3.3)
(f) Underplating required (see 3.3.5)
(g) Luster required, if other than specified (see 3.3.7)
(h) Hydrogen embrittlement relief treatment, if other than specified (see 3.3.8)
(i) Type of chromate treatment required for conversion to type II, if other than specified (see 3.3.9)
(j) Type of phosphate treatment required for conversion to type III, if other than specified (see 3.3.10)
(k) Thickness of coating, if other than specified (see 3.4.1, 3.4.1.1 and 3.4.1.2)
(l) Control record requirement (see 4.5.1.1)
(m) Preproduction control examination (see 4.4.1)
(n) Number of samples for destructive testing (see 4.5.2.3)
(o) Whether corrosion resistance test is required (see 4.5.2.3.3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

6.2.1 Other data. The manufacturer of the basis metal parts should provide the plating facility with the following data:

(a) Hardness or ultimate tensile strength of steel parts (see 3.3.1 and 3.3.8)
(b) Heat treatment for stress relief whether it has been performed or is required (see 3.3.2)
(c) Tensile loads required for embrittlement relief test, if applicable (see 4.6.4).

6.2.2 Specimens. The manufacturer of the basis metal parts should provide the notched tensile specimens for the preproduction testing and lot sampling inspection.

6.3 Limitations. Cadmium platings should not be used on parts for space applications or on parts which in service reach a temperature of 233°C (450°F) or higher or come in contact with other parts which reach those temperatures. MIL-S-5002 contains additional warning where the cadmium plating shall not be used for weapons systems applications.

6.4 Stress relief. There is a hazard that hardened and tempered cold-worked or cold-straightened steel parts may crack during cleaning and plating. Such parts should have a suitable heat treatment for stress relief prior to cleaning and plating (see 3.3.2).

6.5 Type II temperature limitations. Chromate treatments (type II) should not be used on plated parts that will not be painted and which will be continuously exposed to temperatures in excess of 66°C (150°F) or intermittently exposed for short periods to temperatures of approximately 131°C (300°F) or more. However, these treatments may be used to prevent finger marking and corrosion which may occur at room temperature during assembly and storage.

6.6 Type II handling precaution. Chromate treatments (type II), which involve only dipping in chemical solutions, normally require a sufficient period of drying, approximately 24 hours at 21° to 32°C (70° to 90°F), to render the parts suitable for handling without damage to the coating while in gelatinous forms; and it is important with such coatings that the workmanship be such that the coating is not excessively damaged while wet. Type II platings should be dry for 24 hours before painting.

6.7 Reactivation. Surfaces of cadmium plating intended for conversion to type II and which have become passive as a result of the baking operation specified in 3.3.8 may be reactivated by brief immersion in dilute acid. If, for example, the chromating solution is acidified with sulfuric acid, then the reactivating solution should be 1 part sulfuric acid (sp. gr. 1.83) by volume added to 99 parts water, or for further example, if the chromating solution is acidified with hydrochloric acid, then the activating solution should be 1 part hydrochloric acid (sp. gr. 1.16) by volume added to 99 parts water. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

duration of immersion should be as brief as is consistent of the nature of the work. For example, a perforated container of barrel-plated parts would be expected to be reactivated in approximately 15 seconds and separately racked items in approximately 5 seconds. The surfaces should be activated as soon as possible following baking operation and should be handled carefully to avoid contamination by dirt or grease.

6.8 Toxicity. Cadmium, because of its toxicity, should not be employed as a plating for any object intended for use as a food container, cooking utensil or for any object likely to come in contact with food. Cadmium plated sheets and any other structural shapes which may be subjected to heat from welding, brazing or soldering operations should be so labeled because of the danger from poisonous vapors during operations.

6.9 Packaging limitations. Cadmium plated articles should not be packed in non-ventilated containers, either together or in contact with electrical equipment, because of the danger of deleterious effect on cadmium plating from unstable organic electrical insulation. In addition to organic electrical insulation, phenolic resinous substances and other containing unsaturated carbon-to-carbon linkages, such as oil paint and impregnated paper, etc., cause an abnormal attack on cadmium by setting free, in the presence of moisture, formic acid, butyric acid, etc. Corrosion of cadmium coatings and steel basis metal has been noted when cadmium plated articles have been packaged in direct contact with container materials such as wood or cardboard. Corrosion has been especially severe if the container materials have become wet or have been stored under conditions of high humidity.

6.10 Changes from previous issue. Asterisks are not used this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

MILITARY INTEREST:

Custodians

Army - MR
Navy - AS
Air Force - 20

Review Activities

Army - AL, AR, ER
Navy - EC, OS
Air Force - 70, 99
DLA - IS(E)

User Activities

Army - AT, ME, MI
Navy - MC, YD

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
COM - NBS

PREPARING ACTIVITY:

NAVY - AS

DOD Project MFFP-0247

Orders for this publication are to be placed with the Administration, acting as an agent for the Superintendent of Documents. See S... f this specifciation referenced herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

TABLE I. Stress relief and embrittlement relief requirements.

Type of part	Strength level (PSI) Hardness (RC)	Before plating Temperature for stress relief	Before plating Time for stress relief	After plating Time between plating and baking	After plating Temperature for baking	After plating Time for baking
Carburized parts and other parts tempered below 400 °F	---- RC 55 and over	275°F	5 hours	Shall not exceed 4 hours	275°F ± 25°F	23 hours minimum
All other steel parts	Above 160 ksi (Above RC 36)	Between 350 and 400°F	4 hours minimum	Shall not exceed 4 hours	375 ± 25°F	23 hours minimum
	160 ksi and below (RC 36 and below)			Shall not exceed 4 hours	375 ± 25°F	4 hours minimum

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

TABLE II. Thickness.

Class	Thickness	
	Inch, minimum	Approximate equivalent minimum thickness, micrometers 1/
1	0.0005	13
2	0.0003	8
3	0.0002	5

1/ 0.001 inch = 1 mil = 25.4 micrometers (microns)

TABLE III. Production control tests and specimens.

Test	Applicable type	Paragraph		
		Requirement	Specimen preparation	Test method
Thickness	I, II, III	3.4.1, 3.4.1.1 and 3.4.1.2	4.5.3 and 4.5.3.1 1/	4.6.1
Adhesion	I, II, III	3.4.3	4.5.3 and 4.5.3.1 1/	4.6.3
Corrosion resistance	II	3.4.4	4.5.3 and 4.5.3.2 1/	4.6.2
Hydrogen embrittlement	I, II, III	3.3.8	4.5.1.3.1, 4.5.3 and 4.5.3.3	4.6.4

1/ Standard alloy steel shall be used for production control specimens. Alloy steels such as AISI or SAE numbers 4130, 4135, 4140, 4145, 4340, 8645, 8740 and steels conforming to MIL-S-5000 shall be used. Any alloy from the above group is suitable for selection, and shall be at the option of the supplier.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

TABLE IV. Sampling for visual examination and nondestructive tests.

Number of items in lot	Number of items in samples (randomly selected)	Acceptance number (maximum number of sample items nonconforming to any test)
15 or less	7 1/	0
16 to 40	10	0
41 to 110	15	0
111 to 300	25	1
301 to 500	35	1
501 and over	50	2

1/ If the number of items in the inspection lot is less than 7, the number of items in the sample shall equal the number of items in the inspection lot.

GSA DC-8911601297

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.